

Project Completion Report

Project Name: Blockchain voting system **URL:**

<https://projectcatalyst.io/funds/11/cardano-use-cases-concept/blockchain-voting-system> **Project Number:** 1100039 **Project Leads:** David Tacuri

1. Project Description

The Blockchain Voting System project was developed as a proof of concept aimed at exploring the practical application of blockchain technology in electoral processes, specifically in the management and traceability of scrutiny records.

The main focus was not to replace existing electoral systems, but to strengthen them by incorporating an additional layer of audit, integrity, and transparency. To achieve this, a hybrid architecture was designed that combines:

- Traditional systems (backend in C#, APIs, databases)
- Blockchain (Cardano) as a verification layer
- Decentralized storage (IPFS) for digital evidence

Within the system flow, critical processes such as scanning, digitization, and quality control are executed off-chain, while the blockchain is used to record hashes, states, and metadata, ensuring that any alteration can be detected.

This approach balances performance, security, and decentralization, making it viable for real-world environments.

2. Objectives

General Objective To design and validate a blockchain-based system that improves transparency, traceability, and integrity in electoral processes.

Specific Objectives

- Implement a complete traceability mechanism for records, from their digitization to final validation
- Ensure data integrity through the use of cryptographic hashes
- Integrate the Cardano blockchain with a traditional backend using cardano-cli and metadata
- Use IPFS to store images and evidence, reducing the load on the blockchain
- Design and validate a hybrid architecture (on-chain / off-chain) suitable for high-performance systems
- Evaluate the technical and operational viability of this solution in real contexts

3. Methodology and Activities

The project was developed following an iterative approach, focused on validating hypotheses through a Proof of Concept.

Planning

- Definition of the system architecture based on a hybrid model
- Identification of the critical processes of the electoral system:
 - Scanning of records
 - Digitization of results
 - Quality control
- Selection of technologies:
 - Cardano (blockchain)
 - IPFS (storage)
 - Backend in C#

Implementation

- Development of APIs and backend services to manage the flow of records
- Integration with Cardano using cardano-cli for:
 - Transaction construction
 - Signing
 - Submission
 - Metadata registration
- Generation of hashes of the records and storage of images in IPFS
- Definition of a controlled state flow for each record, ensuring that stages cannot be skipped

Testing

- Execution of multiple transactions on the Cardano network (Preview testnet)
- Validation of the complete traceability of the process:
 - Scanning → Digitization → Quality control
- Verification of data integrity through hash comparison
- Observation of behavior in terms of latency and confirmation

Evaluation

- Analysis of system performance against real requirements
- Identification of technical limitations (latency, costs, data volume)
- Evaluation of the hybrid architecture as a viable solution
- Documentation of results, lessons learned, and opportunities for improvement

4. Results

The project achieved:

- Development of a functional Proof of Concept capable of registering and verifying the complete cycle of a record
- Successful integration between:
 - Traditional backend
 - Blockchain (Cardano)
 - IPFS
- Implementation of a verifiable traceability system, where each stage of the process is recorded and can be audited
- Validation that blockchain is effective as a mechanism for:
 - Audit
 - Integrity
 - Verification

Additionally, it was confirmed that:

- Off-chain systems are indispensable for performance
- Blockchain must be used strategically, not as the main processing system

5. Challenges and Solutions

Blockchain Latency Transaction confirmation time on Cardano is not immediate.

Solution: Use blockchain as an audit layer, not for real-time operations.

Data Storage Registering images or detailed data directly on the blockchain is costly and inefficient. **Solution:** Store images and detailed data in IPFS and register only hashes on the blockchain.

System Performance Electoral processes require high speed and concurrency.

Solution: Keep critical processing outside the blockchain (off-chain).

Integration Complexity The interaction between backend, blockchain, and IPFS increases complexity. **Solution:** Modular design and clear separation of responsibilities.

6. Lessons Learned

- Blockchain does not replace existing systems; it complements them
- The balance between on-chain and off-chain is fundamental
- Latency limits its use in real-time processes
- Security is multi-layered (not only blockchain)
- Traceability is the greatest value contributed by the system
- Design must consider costs and scalability from the beginning
- Proof of Concept are essential to validate hypotheses

7. Recommendations

- Integrate Layer 2 solutions such as Hydra to improve scalability
- Explore technologies such as Midnight for privacy of sensitive data
- Optimize transaction usage to reduce costs
- Improve user interfaces for traceability visualization
- Implement pilots in real environments (public institutions)

8. Conclusion

The project demonstrates that blockchain technology can significantly improve transparency, traceability, and trust in electoral processes.

However, its effective implementation requires a hybrid approach, where traditional systems manage performance and blockchain acts as an independent verification layer.

This model represents a solid foundation for future implementations in real governmental environments.